

CSA1304-THEORY OF COMPUTATION LAB EXPERIMENTS

2. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with 0 and end with 1

AIM

Write a C program to simulate a **Deterministic Finite Automata (DFA)** for the language representing strings that **start with '0' and end with '1'**.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int len;

    printf("Enter a binary string: ");
    scanf("%s", str);

    len = strlen(str);

    if(len > 0 && str[0] == '0' && str[len - 1] == '1')
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

Output

Clear

Enter a binary string: 001101
String Accepted

=== Code Execution Successful ===